CURRICULUM VITAE

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NOW

I simulate microscope optics with the Physics of Life group at TU Dresden for the summer, then go back to Barcelona for my fourth year at UPC. I want to work where physical modeling and machine learning meet.

EXPERIENCE

SINCE 2026

Contributor, Towards Data Science

Long-form articles since March 2026, on simulation, high-performance computing and machine learning. Most of them come out of the projects below.

2026

Research intern, Physics of Life, TU Dresden

Simulating electromagnetic propagation through specimen and objective, as virtual instrumentation for learned brightfield imaging.

JUN–AUG 2025

Machine learning intern, MLCode

Built an automated framework for evaluating and benchmarking retrieval-augmented generation systems.

SINCE 2022

Volunteer mentor

Computer science and robotics workshops: Linux, 3D printing, programming.

2022–2023

Robotics teacher, Punt Multimèdia

Taught Arduino, micro:bit, and 3D printing and modeling.

EDUCATION

SINCE 2023

BSc Data Science & Engineering, UPC

Entering the fourth year of five. 8.75/10 average across both degrees.

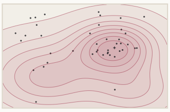
SINCE 2023

BSc Physical Engineering, UPC

Taken in parallel with the data science degree rather than after it, on the CFIS dual-degree program.

SELECTED PROJECTS

2026

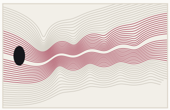


Aerodynamic optimization of a Formula 1 front wing

A surrogate model searched front-wing geometries on a fixed budget of solver runs rather than calling CFD at every step. The wing came out about 15% better on lift-to-drag.

Python · OpenFOAM · SLURM · MareNostrum V

2026

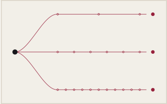


A physics-informed network for vortex shedding

A PINN with the Navier-Stokes residual inside its loss, trained against a high-fidelity OpenFOAM run, reproducing Kármán vortex shedding. A model fit to the data alone settles into a steady wake instead.

Python · PyTorch · OpenFOAM

2026

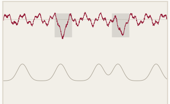


Persona adaptation in language models

A system prompt against three formats of fine-tuning data on Qwen3-4B-Instruct, all with LoRA at r=16, so the differences belong to the data.

Python · PyTorch · Transformers · LoRA

2025

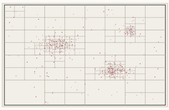


Predicting fetal distress from cardiotocography

Built with Hospital Sant Joan de Déu in a team of nine. Separate encoders over CTG traces, tabular clinical context and ultrasound, at around 0.78 AUC.

`Python · PyTorch`

2025

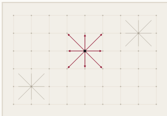


Parallel N-body simulation

Direct $O(n^2)$ summation and a Barnes-Hut $O(n \log n)$ tree code in C++ with OpenMP. At N = 10,000 direct summation ran about 33× faster on 112 threads than on one; the tree code never got past roughly 7×.

`C++ · OpenMP`

2025

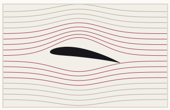


Parallel lattice Boltzmann solver

A D2Q9 solver with BGK collision and a Zou-He inlet in around 200 lines of C++: flow past a cylinder on a 400×400 lattice. Memory-bound, so how you move the data dominates the scaling.

`C++ · OpenMP · MareNostrum 5`

2025

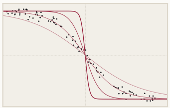


Fixed-wing bird flight simulation

A 3D Navier-Stokes solver written from scratch in NumPy, run on a bird's wing modeled in Blender and checked against OpenFOAM's icoFoam.

`Python · NumPy · OpenFOAM · Blender`

2024



Fermi-Dirac statistics by Monte Carlo

A Markov chain over the microstates of a fermion system, recovering the Fermi-Dirac occupancy numerically rather than analytically.

`Python · NumPy`

Figures are schematic illustrations of each problem, not results.

SKILLS

LANGUAGES	Python · C++ · R · Haskell · MATLAB
SCIENTIFIC	OpenFOAM · OpenMP · NumPy · PyTorch · Blender · CAD
METHODS	CFD · HPC & SLURM · Physics-informed ML · Monte Carlo · Surrogate modeling
SPOKEN	Catalan (native) · Spanish (native) · English (professional) · Arabic (beginner)

AWARDS

2024, 2025	Participant, Datathon FME
2025	Physics summer school, University of Ljubljana
SINCE 2023	Cum laude in several subjects
2023	Admitted to CFIS on a grant, to read the two degrees at once. It takes a small cohort each year by its own entrance exam in mathematics and physics
2019–2023	Graduated from high school with honors, and represented it at the Barcelona *Mostra de Recerca Jove*, after UPC's summer courses in algorithms and programming